

# Did Inequality Fall in Chile?

## Income, Wealth, and Survey Disagreement, 2007–2024

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### Abstract

Chile's widely reported twenty-first century decline in income inequality appears in several international databases, but many of those series ultimately draw on the same underlying survey: the National Socioeconomic Characterization Survey (CASEN). This paper asks whether recent compression also appears in a different survey system and whether income and wealth tell the same story through 2024. I harmonize six waves of the Household Financial Survey (EFH), retain all 30 implicates and the released survey designs, and construct the closest feasible comparison with urban CASEN households. EFH records modest income compression through 2017, an exceptional fall in 2021, and a return to approximately its 2017 Gini by 2024: 0.519, 0.439, and 0.523, respectively. Its income Gini is 0.030 lower in 2024 than in 2007, with a conservative interval that also admits little net change. CASEN instead moves from 0.482 in 2017 to 0.464 in 2024, widening the EFH-minus-CASEN gap from 0.037 to 0.059, although the direct difference in 2017–2024 changes remains imprecisely estimated. EFH signed non-pension wealth inequality rose between 2021 and 2024, from 0.698 to 0.766. Income-component accounting associates the exceptional 2021 compression with temporary changes in labor and other autonomous income and with public transfers taking a much larger share of measured EFH income. The reversal remains in the 1,801-household 2024 survivor domain under both common-2021 and common-2024 weights, so refresh-sample composition is not a sufficient explanation. The results leave the historical CASEN decline intact while showing that its continuation into the 2020s depends on the survey system and on whether income or wealth is examined. The disagreement is informative without making either survey a stand-in for the unobserved true distribution.

**Keywords:** Chile; income inequality; wealth inequality; household finance survey; public transfers; survey disagreement

**JEL classification:** C81; D31; E01

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I thank the participants of my research seminar, especially Eric Monnet and Jérôme Bourdieu, for their helpful suggestions as well as Xavier Jara-Tamayo for accepting to act as referee. This thesis would not have been possible without the patient support and guidance provided by my supervisor, Facundo Alvaredo. All errors are my own.

# 1 Introduction

Chile appeared to be moving against one of the defining distributional concerns of the early twenty-first century. While rising concentration returned to the center of economic research, household-survey estimates placed Chile’s Gini near the Latin American peak around 1990 and roughly ten points lower by the middle of the 2010s (Alvaredo et al., 2025). The *Encuesta de Caracterización Socioeconómica Nacional* (CASEN; National Socioeconomic Characterization Survey), the country’s principal survey for poverty and income distribution, supplies the most continuous evidence behind that account. Together these estimates point to substantial compression, though its timing and magnitude vary with the series. Whether that compression persisted beyond the 2010s is less settled. The central puzzle is whether the decline remains visible through 2024 when household resources are measured in a different survey system—and whether wealth, rather than income alone, points in the same direction.

Survey-based inequality is sensitive to what is measured and how the data are processed. A household total, household income per capita distributed among persons, equivalized disposable income, and equal-split national income assign different resources to different units. Nonresponse corrections, imputed rent, upper-tail treatment, and population coverage introduce further variation. Studies of secondary inequality databases and Latin American microdata show that these choices can change both country rankings and comparisons over time (Atkinson and Brandolini, 2001; Székely and Hilgert, 2007; Alvaredo et al., 2025). Incomplete coverage at the top creates another source of uncertainty (Blanchet et al., 2022). The underlying microdata are therefore as important as the published series. For Chile, the Socio-Economic Database for Latin America and the Caribbean (SEDLAC), the Economic Commission for Latin America and the Caribbean (ECLAC), and the World Bank’s Poverty and Inequality Platform (PIP) ultimately process CASEN observations. The Luxembourg Income Study (LIS) and the Organisation for Economic Co-operation and Development (OECD) Income Distribution Database do so as well. Their agreement shows that the historical decline survives several treatments of CASEN; it cannot establish that the trend is robust to different underlying household-survey microdata.

The *Encuesta Financiera de Hogares* (EFH; Household Financial Survey) offers a counterpoint. Designed around household balance sheets, EFH uses a wealth-related oversampling frame and multiple imputation for missing financial items. Its income aggregate differs from CASEN’s official construct, but the released files allow several major comparison choices to be aligned: survey year, urban coverage, household unit, and the treatment of imputed rent. Even then, the surveys support different recent paths. The puzzle is therefore simple to state: how can an inequality decline reproduced across prominent household-survey databases coexist with EFH evidence of only modest net compression from 2007 to 2024 and no continuing decline after 2017?

This paper asks whether Chile’s survey-measured decline persists in a different survey system and what income composition and the EFH survivor sample reveal about the divergent post-2017 paths. I estimate the recent trend supported by each source under an explicit estimand and use the released data to examine parts of their disagreement. Judging which survey is closest to an

unobserved true distribution lies outside the design.

Three findings organize the answer. First, EFH records modest income compression through 2017, an exceptional decline in 2021, and a return to approximately its 2017 level in 2024. The corresponding household-income Ginis are 0.519, 0.439, and 0.523. The 2024 estimate is 0.030 below its 2007 value, but the endpoint is too imprecisely estimated to distinguish a modest net decline from little net movement. Second, CASEN’s closest household-income Gini moves from 0.482 in 2017 to 0.464 in 2024, while the EFH-minus-CASEN gap widens from 0.037 to 0.059. Third, EFH signed non-pension wealth inequality rose between 2021 and 2024, from 0.698 to 0.766, its highest point estimate in the six-wave series.

Two diagnostics give the income reversal more context. Public transfers’ share of measured EFH effective income rises from 0.8 percent in 2017 to 13.5 percent in 2021 and recedes to 3.1 percent in 2024. Exact component accounting locates the rest of the swing in labor and other autonomous income. Among the 1,801 households reinterviewed in 2024, the reversal persists under both common-2024 and common-2021 weighting systems. Refresh-sample composition is therefore not a sufficient explanation, although attrition and within-household measurement remain unresolved.

The paper contributes the 2024 EFH endpoint to the evidence on Chilean inequality, adding a new income observation, a wealth increase, and growing disagreement with CASEN. Related work integrates pensions, macro balance sheets, and upper-tail corrections through 2021 ([Castro Nofal et al., 2026](#)). That complementary construction addresses a broader wealth concept, whereas the present comparison retains the released EFH estimand. Multiple-imputation inference secures the reported uncertainty; component accounting and the survivor domain then probe two concrete explanations for the reversal. Neither exercise identifies the cause of the disagreement.

The remainder of the paper proceeds as follows. Section 2 establishes the Chilean decline, audits its source lineage, and explains what EFH adds. Section 3 defines the data, estimands, and inference. Section 4 reports EFH income and wealth trends. Section 5 characterizes the post-2017 disagreement through the closest-concept comparison, income components, and the survivor domain. Section 6 closes with what the evidence implies for the claim that Chilean inequality has continued to fall through 2024.

## 2 Chile’s inequality decline and its evidentiary lineage

### 2.1 The established decline and the measurement problem

CASEN supplies Chile’s principal repeated record of household income, poverty, and monetary subsidies. Its continuity helps explain why declining inequality became the default account of the first two decades of this century. Chilean studies describe an increase during the 1990s followed by a marked fall between 2000 and 2011, while regional harmonizations carry the decline into the middle of the 2010s ([Parro and Reyes, 2017](#); [Alvaredo et al., 2025](#)). In their review of Latin

American inequality, [Alvaredo et al. \(2025\)](#) place Chile’s Gini about ten points below its value around 1990 and characterize the movement after 2015 as a mild increase.

The economic literature has placed labor earnings at the center of that decline. For 2000–2011, decompositions emphasize the expansion of tertiary education, a falling wage premium for skills, and the resulting compression of labor income; more progressive public transfers also contributed, but on a smaller scale than labor income in the Chilean account ([Parro and Reyes, 2017](#)). Recent CASEN work extends the study of education and labor-income dispersion through 2024, while documenting weaker real earnings for tertiary graduates after 2015 ([Rodríguez Tapia, 2026](#)). These explanations concern changes measured in CASEN. They motivate the component analysis below while leaving open whether the same movements appear in a survey built for household finances.

Figure 4 of that review motivates the lineage audit developed here. It places SEDLAC, ECLAC’s newer survey-only series, PovcalNet/PIP, and LIS inside a preferred band of four microdata-based household-survey series. Across the period beginning around 1990, those lines, supplemented by more fragmented published series, trace a clear medium-run decline in Chile and reach their reported troughs in 2015. Their levels differ, but each comparison distributes household per-capita income among persons. Much of the remaining variation comes from how a common upstream survey is processed rather than from four independent samples of Chilean households.

The supplement plots that historical benchmark in one frozen CASEN-derived PIP series. PIP applies its own welfare construction and population conventions, so the plot documents the prevailing long-run account without adding an independent survey or a direct level comparison with EFH.

Evidence outside that survey band complicates the size and timing of the decline. Tax-based studies show that business income, retained earnings, and upper-tail treatment materially affect measured concentration ([Fairfield and Jorratt De Luis, 2016](#); [Flores et al., 2020](#)). Adjusting CASEN with national accounts and administrative records can also flatten the 2003–2017 trend, depending on the income concept and correction used ([Larrañaga et al., 2022](#); [Ministerio de Desarrollo Social y Familia, 2026](#)). Distributional national accounts widen the resource boundary further by allocating income absent from surveys ([Piketty et al., 2018](#); [Alvaredo et al., 2022](#)). These results qualify the survey narrative rather than supply another independently sampled household-income path.

## 2.2 One survey lineage, several published series

Source lineage records the upstream microdata as well as the downstream publisher. A recent global audit reaches a closely related conclusion: apparent plurality across inequality databases partly reflects database proliferation and overlapping genealogies rather than independent measurement ([Ceriani and Verme, 2026](#)). That study provides useful corroboration at a different scale. The audit here traces prominent Chilean observations back to their household surveys

and separates shared microdata from shared methodology. SEDLAC constructs harmonized current household income and commonly reports household per-capita income distributed among persons; its Chilean input is CASEN ([CEDLAS and World Bank, 2025](#)). World Bank PIP draws on SEDLAC for Latin America and applies additional treatment, so the two series need not be numerically identical ([World Bank, 2023](#)). ECLAC's newer series is computed from its Household Survey Data Bank (BADEHOG), whose Chilean files are CASEN ([Economic Commission for Latin America and the Caribbean, 2025](#)). LIS's Chilean income files are harmonizations of CASEN ([LIS Cross-National Data Center, 2018](#)). OECD's Income Distribution Database also lists CASEN as the Chilean source, but distributes equivalized disposable cash income among persons and excludes imputed rent ([OECD, 2026](#)).

WIID plays a different role. It is a secondary database and companion-series producer that gathers and standardizes estimates from national statistics, SEDLAC, LIS, OECD, and other providers ([UNU-WIDER, 2025](#)). The Standardized World Income Inequality Database (SWIID) similarly uses a model to standardize observations toward a common LIS-based concept ([Solt, 2020](#)). Both can broaden historical coverage or support cross-country analysis, but neither turns a published Chilean estimate into a new household survey. Conversely, WID combines survey, administrative, fiscal, and national-accounts information, and UBS or Credit Suisse combines surveys, balance sheets, and models to estimate adult wealth. These constructions bring distinct information and methods to the question, though their estimands lie farther from the household-level EFH measure. The *World Inequality Report 2026*, for example, describes Chilean inequality as having declined noticeably since 2014, but its 2024 series distributes pre-tax national income among individuals rather than reproducing a household-survey Gini ([World Inequality Lab, 2026](#)).

Chile also has independent microdata that deserve mention. The National Statistics Institute's Supplementary Survey of Income (ESI) is attached to the National Employment Survey and is now focused on labor income; it no longer publishes the former household-income indicators. The University of Chile's long-running Employment and Unemployment Survey supports valuable historical income work but represents Greater Santiago rather than the national urban population ([Instituto Nacional de Estadísticas, 2025](#); [Sanhueza and Mayer, 2011](#)). The National Statistics Institute's Household Budget Survey (EPF) independently measures household income and expenditure, but covers regional capitals and their conurbations, runs roughly every five years, and does not supply a 2024 endpoint under the same income definition ([Instituto Nacional de Estadísticas, 2022](#)). These surveys are informative checks on particular populations or resources; none furnishes the conceptually stable national- urban household-total-income sequence needed here. I therefore use EFH as the main survey counterpoint to CASEN.

The wealth lineages run differently. OECD's Wealth Distribution Database identifies the Survey of Household Finances and the Central Bank of Chile as its Chilean source. The Luxembourg Wealth Study likewise states that its CL07, CL14, CL17, CL21, and CL24 files are based on EFH ([OECD, 2025](#); [LIS Cross-National Data Center, 2025](#)). Treating those series as

independent confirmations of EFH would count the same upstream survey more than once.

Figure 1 summarizes the sources audited in this paper. Arrows indicate documented upstream inheritance, not identical methods or published numbers. Independence and comparability are separate dimensions: a series may be closely comparable but derivative, or independently constructed while estimating a different population and resource.

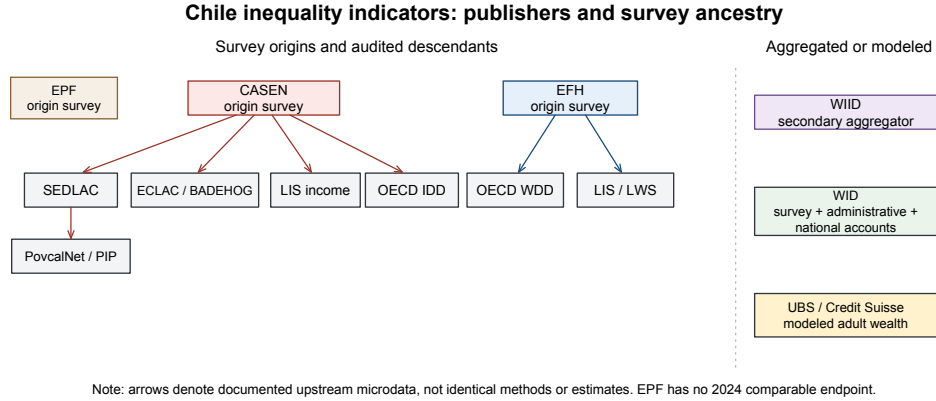


Figure 1: Source lineage of the series audited in this paper

Agreement across the CASEN family therefore supports a precise conclusion: Chile’s historical decline is robust to several treatments of CASEN. Among the sources audited here, however, a comparable national household-total-income series built from independent microdata has not reproduced the full path. EFH begins only in 2007, so it cannot adjudicate the early decline, but it can test how confidently the more recent direction travels across survey systems. The audit accordingly counts transformations and independent measurements separately, without discounting the information contained in either.

### 2.3 What EFH adds

EFH is a different instrument. It is designed around household finances and balance sheets, uses oversampling tied to a wealth proxy, and releases multiple imputations for item nonresponse. Its unit is the household and its national waves represent urban households. The effective-income measure used here adds labor income, pensions, public subsidies, and other income at the household level while excluding imputed owner rent. CASEN’s official monetary-income measure instead aggregates corrected person-level autonomous income and cash subsidies under the ministry’s procedures. The closest comparison removes imputed rent from both, but the questionnaires and correction systems remain different. Comparing their Gini levels as if the income concepts were identical would therefore overstate precision. These same differences make

EFH useful as an independent survey check rather than another treatment of CASEN.

EFH has been used primarily to study debt, financial fragility, and wealth. Early work documented the distribution of assets, debts, and income, while recent research adds pensions, national balance sheets, and upper-tail corrections through 2021 (Cox et al., 2006; Martínez and Uribe, 2017; Banco Central de Chile, 2007–2024; Castro Nofal et al., 2026). This paper deliberately retains the narrower released survey concepts. Their value lies in a consistent six-wave account, the new 2024 endpoint, and the joint observation that recent income and wealth do not move toward an unambiguous compression narrative.

## 2.4 The unsettled endpoint and pandemic comparability

The latest observation changes how the recent record is read. A series that is lower in 2024 than in 2017 supports continued compression. A series that falls exceptionally in a pandemic round and then returns to its earlier level supports temporary compression. Both descriptions can fit parts of Chile’s public record, but they support different conclusions about the direction through 2024.

The 2021 EFH and 2020 CASEN are also unusual observations. EFH fieldwork was largely remote during emergency transfers and labor-market disruption, while CASEN used an exceptional, primarily telephone collection design (Cerletti et al., 2024; Ministerio de Desarrollo Social y Familia, 2022). Income, who responded, and how respondents were contacted changed together. No design here separates those margins. The main comparison therefore uses same-year 2011, 2017, and 2024 observations. CASEN 2020 and 2022 describe its own higher-frequency path; they are not mechanical matches or mode corrections for EFH 2021.

The three inferential contrasts emphasized below are the 2017–2021 EFH income fall, the 2021–2024 income reversal, and the 2021–2024 wealth increase. The endpoint, component account, and survivor-domain exercise are descriptive diagnostics. They can narrow interpretations without identifying why the survey systems disagree.

## 3 Data, estimands, and inference

### 3.1 EFH waves and primary resources

The main EFH sequence uses the national-urban waves from 2007, 2011, 2014, 2017, 2021, and 2024. The 2008, 2009, and 2010 Greater Santiago surveys are not inserted into the national series. The statistical unit is the household and the weight is the provider’s household expansion factor.

The primary income resource is effective monthly household income excluding imputed owner rent, named `ytothef` in the two earliest files and `ytotef` thereafter. Total income including imputed rent is a sensitivity. The primary wealth resource is signed non-pension net wealth, defined as the provider-consistent asset total less the debt total. Negative values remain negative.

A zero-capped Gini is reported only to show the effect of that normalization. “Non-pension” means that mandatory retirement balances are excluded; constructing a comparable pension-wealth measure would require auxiliary records and assumptions that are not available consistently across the six released waves. Macro scaling, Pareto corrections, and a top-one-percent series are also outside scope.

Table 1 summarizes the six-wave analytical sequence. The supplement provides a parallel EFH–CASEN crosswalk with field periods, collection modes, income reference periods, and response rates wherever the released documentation reports them.

| Wave | Households | Fieldwork and mode                                                            | Income reference                  | Response rate |
|------|------------|-------------------------------------------------------------------------------|-----------------------------------|---------------|
| 2007 | 3,828      | Nov. 2007–Jan. 2008; in person                                                | Current monthly; labor last month | n.r.          |
| 2011 | 4,059      | Sep. 2011–May 2012; in person                                                 | Current monthly; labor last month | n.r.          |
| 2014 | 4,502      | Jul. 2014–Feb. 2015; in person                                                | Current monthly; labor last month | n.r.          |
| 2017 | 4,549      | 20 Jun.–5 Dec. 2017; in person                                                | Current monthly; labor last month | n.r.          |
| 2021 | 4,400      | 4 Sep. 2021–31 Mar. 2022; precontact in person, interview mainly by telephone | Current monthly; labor last month | n.r.          |
| 2024 | 4,649      | 22 Jul. 2024–13 Jan. 2025; in person                                          | Current monthly; labor last month | 71.5%         |

Table 1: EFH analytical sequence and fieldwork. “n.r.” means that an overall response rate was not reported in the reviewed wave document; target-attainment rates are not substituted for response rates.

Every nonlinear statistic is estimated separately in all 30 completed data sets and pooled with Rubin’s rules. The 2017, 2021, and 2024 intervals use all 1,000 provider replicate weights; the earlier releases lack those weights and use a fixed-seed bootstrap within the released income strata, so their intervals remain qualified. Signed wealth retains negative balance sheets, and bottom-50 and top-10 shares use fractional population boundaries. A change is called robust only when it survives both a covariance-agnostic bound for the two wave estimates and Holm adjustment across the registered adjacent-wave family. The supplement gives the estimators, proofs, and validation results.

### 3.2 CASEN comparison

CASEN rounds from 2011, 2017, 2020, 2022, and 2024 are reduced to one record per household head and restricted to urban households. The resource is corrected monetary household income excluding imputed rent. Estimates use the released expansion factor, pseudo-stratum, and pseudo-PSU fields, together with the 2011 finite-population correction where available ([Ministerio de Desarrollo Social y Familia, 2011–2024](#)). Taylor linearization supplies the main Gini intervals. Deterministic stratified delete-one-PSU jackknife replicates supply uncertainty for the nonlinear component statistics.

The direct EFH and CASEN comparison remains limited to 2011, 2017, and 2024. These align survey year, urban scope, household unit, and the exclusion of imputed rent. The income questionnaires and correction procedures are still not identical. Difference intervals treat the surveys as independent and do not include a model for the residual concept or fieldwork gap.

### 3.3 What harmonization removes, and what it leaves behind

The comparison removes four transparent differences: rural coverage, repeated person records for a household resource, imputed owner rent, and unmatched survey years. It does not harmonize how income is elicited or corrected. EFH is organized around household finances and balance sheets, whereas CASEN is organized around poverty, labor, social policy, and living conditions. The respondent, recall period, question order, item nonresponse, top-income coverage, and editing rules can therefore remain different after the final columns have similar labels.

The analysis relies only on released files and published methodological documents. Those materials do not resolve every cross-wave difference in income construction and survey design; questions sent to Banco Central de Chile remain part of the release record rather than an input to the results.

The resulting gap is a difference between two released survey estimands, not a decomposition of survey bias. Separating sampling, questionnaire, response, correction, and mode would require a bridge study that asks both modules of a common household sample or links both surveys to a shared administrative outcome. Here the useful standard is to align what can be aligned, quantify uncertainty from each design, and name the differences that remain.

### 3.4 Income-component accounting

For EFH in 2014, 2017, 2021, and 2024, total effective income is divided into labor income  $y_{labh}$ , public transfers  $y_{subh}$ , and other autonomous income  $y_{penh} + y_{otroh}$ . For CASEN, corrected monetary income is divided into corrected labor income, monetary transfers, and corrected autonomous income less labor income. The derived CASEN transfer component ensures exact add-up; its difference from the released direct subsidy field is 1 or 2 CLP in most rounds and 3 CLP for a negligible share of the revised 2020 file. The discrepancy is retained in the validation record.

Let  $s_{kt}$  denote component  $k$ 's share of mean total income and  $C_{kt}$  its concentration coefficient when households are ranked by total income. Tied totals receive a common weighted midrank. The total Gini is

$$G_t = \sum_k s_{kt} C_{kt}.$$

The change in each component's contribution is divided symmetrically:

$$\Delta(s_k C_k) = \frac{1}{2} \Delta s_k (C_{kt} + C_{k0}) + \frac{1}{2} \Delta C_k (s_{kt} + s_{k0}).$$

The first term is called the share effect and the second the concentration effect. These are accounting terms, not behavioral or causal effects.

### 3.5 Survivor domain and disclosure safeguards

The released 2024 household key links exactly 1,801 reinterviewed households to the 2021 refresh cohort. For this domain, both years first use the 2024 final weight and all 1,000 2024 provider replicates; the exercise is then repeated with common 2021 weights and replicates. Each year’s 30 implicates are estimated separately. The cross-wave imputations were not jointly drawn, so changes again use the covariance-agnostic bound. Baseline income, wealth, household size, macrozone, and stratum composition are compared with the 1,054 households in the 2021 refresh cohort that were not reinterviewed, using 2021 weights.

The linkage produces aggregate estimates only. Household keys and trajectories never enter the outputs, nor do unweighted or small-support cells. The module also checks the linkage, common weights, replicate alignment, and disclosure rules before reporting a survivor-domain result.

The validated samples contain 3,828, 4,059, 4,502, 4,549, 4,400, 4,649 households in the six national-urban waves, respectively. All waves contain 30 implicates, have exactly zero completed-data weighted missingness for primary effective income and net wealth, and pass the prespecified income-accounting tolerance. The 2017, 2021, and 2024 intervals use the provider’s 1,000 replicate weights; earlier-wave intervals remain qualified approximations.

## 4 The EFH path: an exceptional compression and rebound

Figure 2 gives the central result. EFH income inequality falls modestly between 2007 and 2017, drops exceptionally in 2021, and returns to approximately its 2017 value in 2024. Signed non-pension wealth is high throughout and rose sharply between 2021 and 2024.

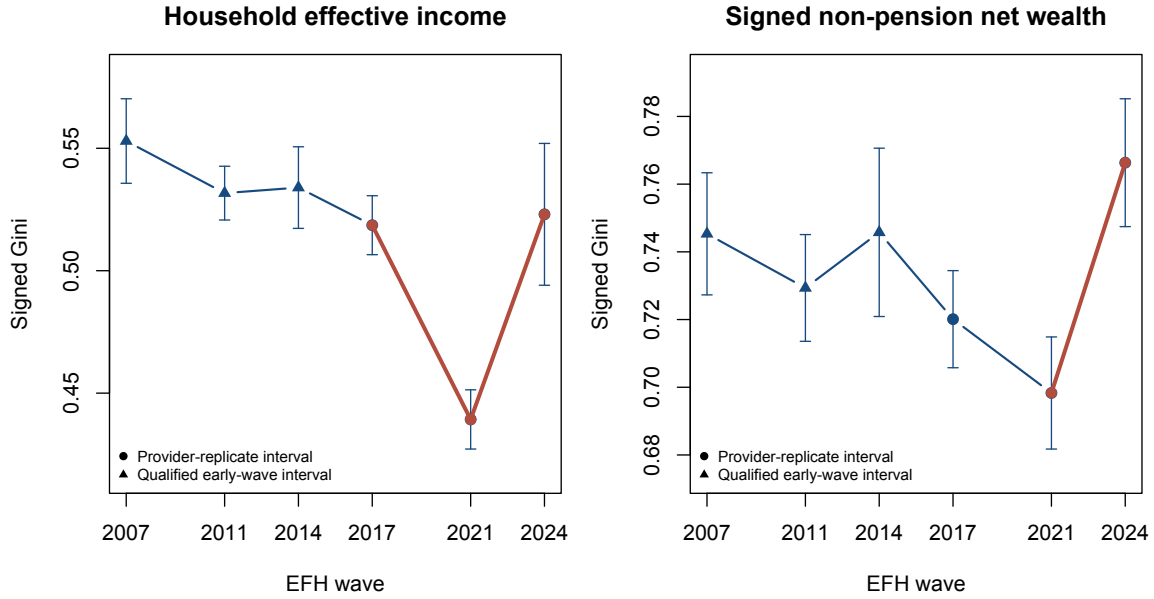


Figure 2: Did inequality fall? EFH income and signed wealth

Household effective-income Ginis (point estimate with 95% interval) were 2007: 0.553 [0.536, 0.570]; 2021: 0.439 [0.427, 0.451]; 2024: 0.523 [0.494, 0.552]. Signed household net-wealth Ginis were 2007: 0.745 [0.727, 0.763]; 2021: 0.698 [0.682, 0.715]; 2024: 0.766 [0.747, 0.785].

Across the full period, the 2024 EFH household effective-income Gini is 0.030 below the 2007 point estimate. The conservative 95% interval for the 2024-minus-2007 difference runs from -0.076 to 0.016, leaving room for both little net movement and a larger decline. The endpoints are therefore too imprecisely estimated to label the series statistically unchanged.

Formal adjacent-wave tests yield a clearer result. Only 3 of the ten Gini changes remain statistically distinguishable from zero after Holm adjustment and under the widest feasible cross-wave interval. The income Gini fell by 0.079 from 2017 to 2021 and rose by 0.084 from 2021 to 2024. The signed wealth Gini rose by 0.068 from 2021 to 2024. All earlier adjacent changes include zero under the conservative interval. These estimates describe repeated cross sections and carry no causal interpretation.

The six-wave shape gives the endpoint its meaning. Compression before the pandemic is modest beside the 2021 drop, and the 2024 estimate closes almost the entire gap with 2017. The most faithful summary is a gradual pre-2017 decline followed by an exceptional trough and a rebound. Across the full period, the endpoint estimates leave room for either a modest net decline or little net movement; within the series, however, the two movements surrounding 2021 are large enough to survive formal adjacent-wave tests.

Those tests are deliberately conservative. The releases do not identify the cross-wave covariance, which may be nonzero because some households are reinterviewed. I therefore allow the covariance that produces the widest feasible interval and adjust for the ten registered adjacent comparisons. Only the 2017–2021 income fall, the 2021–2024 income rebound, and the 2021–2024

wealth increase remain statistically distinguishable from zero. The supplement reports the full level, share, and adjacent-contrast tables.

The 2021–2024 results are not peculiar to the Gini. The income bottom-50 share falls by 0.049 as the top-10 share rises by 0.068; the wealth shares move in the same unequalizing direction. The wealth-Gini increase also remains when negative balances are set to zero. Nor is either headline movement removed by capping each wave at its weighted p99 or p99.5: those caps are recomputed in every implicate and replicate and reported in the supplement as influence checks, not replacements for the uncapped distributions.

Income is a monthly flow, whereas net wealth is a stock of assets less debts. By 2024 the EFH income distribution had lost the compression seen in 2021 while the signed wealth distribution had become more unequal. Taken together, the movements resist a simple account of a uniform decline in measured inequality after 2017.

The wealth result has a deliberately bounded interpretation. Negative-wealth prevalence, bottom-50 and top-10 shares, and the zero-capping sensitivity remain part of the analysis. Zero-capping changes the normalization, but the 2021 to 2024 increase is the headline because 2024 is the new survey information. Pension wealth, macro-balance-sheet scaling, and upper-tail correction could change the level and interpretation. They would answer the broader wealth question studied by [Castro Nofal et al. \(2026\)](#), not the released EFH question here.

The lower tail gives the signed wealth coefficient economic content. A nontrivial share of represented households has liabilities that exceed the included assets, so the bottom-50 wealth share can be negative. Replacing negative values by zero would erase balance-sheet shortfalls and change both the mean used to normalize the Gini and the distances between households. That sensitivity is reported because readers may prefer a nonnegative distributional index, not because zero is a better measurement of debt. The same reasoning is why this paper avoids a top-one-percent claim. In a survey of this size, a few highly weighted observations would dominate both the estimate and its disclosure risk.

Income and wealth also differ in timing. Monthly income can respond quickly to employment and transfers. Wealth combines housing, businesses, financial assets, durable goods, and several forms of debt whose valuations and adjustment speeds differ. A household can therefore return to a more unequal income distribution while the balance-sheet distribution becomes still more unequal. This joint movement makes wealth relevant to the interpretation of recent distributional change, although the analysis stops short of the comprehensive wealth reconstruction already available through 2021.

## 5 Characterizing the post-2017 disagreement

### 5.1 The closest income comparison

Figure 3 places the three same-year EFH comparisons beside the two intermediate CASEN rounds. CASEN rises from 0.482 in 2017 to 0.518 in the 2020 pandemic round, then falls to 0.473 in 2022 and 0.464 in 2024. EFH instead records its trough in 2021 and is back at 0.523 in 2024. The two surveys thus agree that the pandemic period was unusual but disagree about the path from 2017 to the latest common year.

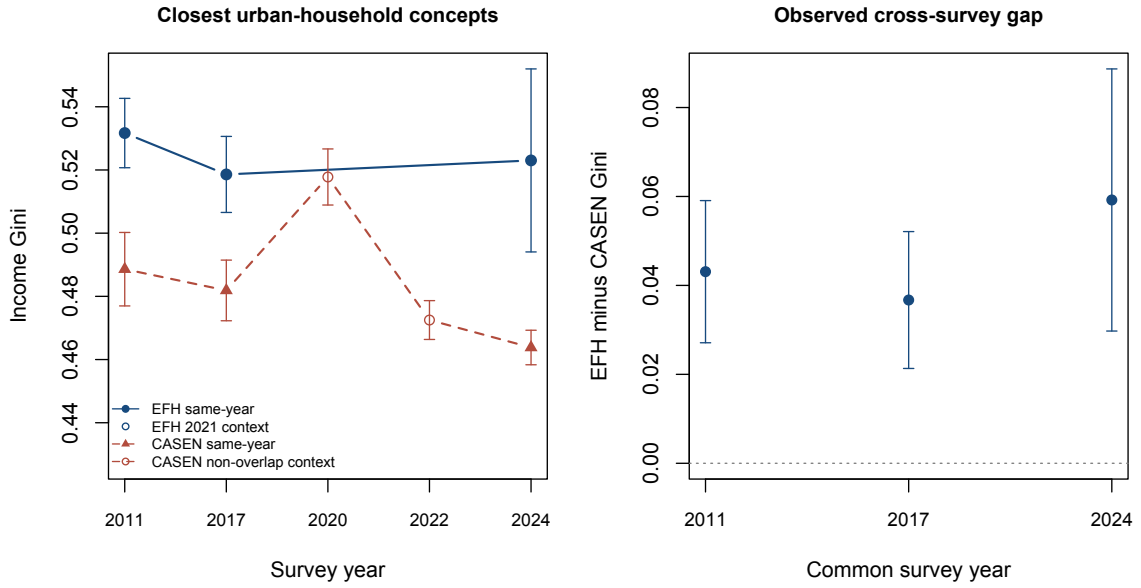


Figure 3: Closest-concept EFH and CASEN income inequality

For the closest available common concept, CASEN urban-household monetary-income Ginis were 2011: 0.489 [0.477, 0.500]; 2017: 0.482 [0.472, 0.491]; 2020: 0.518 [0.509, 0.527]; 2022: 0.473 [0.466, 0.479]; 2024: 0.464 [0.458, 0.469]. The EFH-minus-CASEN income-Gini gaps were 2011: 0.043 [0.027, 0.059]; 2017: 0.037 [0.021, 0.052]; 2024: 0.059 [0.030, 0.089]. All three remain different from zero after Holm adjustment within the Gini comparisons. The intervals account for each survey’s sampling uncertainty, including the CASEN complex design and EFH multiple imputation. They do not account for the residual questionnaire, fieldwork, nonresponse, or income-construction gap and therefore cannot assign the difference to either survey.

Table 2 makes the recent comparison explicit. CASEN’s 2017–2024 decline is distinguishable from zero under the independent-wave calculation, whereas EFH’s small increase is not. The difference between those changes is positive, but its interval includes zero under either variance convention. Thus the evidence establishes different point-estimate paths and a widening level gap, rather than a statistically resolved difference in trends. The same directional contrast appears when income is assigned to represented persons or imputed rent is included, though these two conventions are post-analysis descriptive checks.

| Convention                          | EFH    | CASEN  | Difference in changes |
|-------------------------------------|--------|--------|-----------------------|
| Households, excluding rent          | 0.004  | -0.018 | 0.022 [-0.011, 0.056] |
| Represented persons, excluding rent | 0.001  | -0.019 | 0.020 [-0.012, 0.053] |
| Households, including rent          | -0.003 | -0.022 | 0.018 [-0.014, 0.050] |

Table 2: Income-Gini changes from 2017 to 2024 under alternative conventions

Entries are 2024 minus 2017 Gini changes. The final column reports the EFH change minus the CASEN change with an independent-survey 95 percent interval. Represented-person and including-rent results are post-analysis descriptive checks and do not enter the primary hypothesis family.

The observed gap is not a decomposition of survey bias. The 2024 EFH estimate has wider uncertainty than CASEN because its sample is smaller and its multiple-imputation and replicate variation are material. Even after sampling uncertainty is included, however, the cross-survey gap is positive in all three common years. The remaining explanation could involve questionnaire structure, income correction, top-income coverage, response, fieldwork, or other design features. The present data do not identify their separate effects.

## 5.2 What changed inside household income?

Figure 4 reports each component’s exact contribution to the total-income Gini. The EFH decomposition clarifies both sides of the 2021 episode: labor and other-autonomous contributions contract as transfers take a much larger share of measured income, then those movements largely reverse by 2024. Transfer receipts also become less concentrated among households ranked near the bottom of total income, which offsets part of the equalizing share shift.

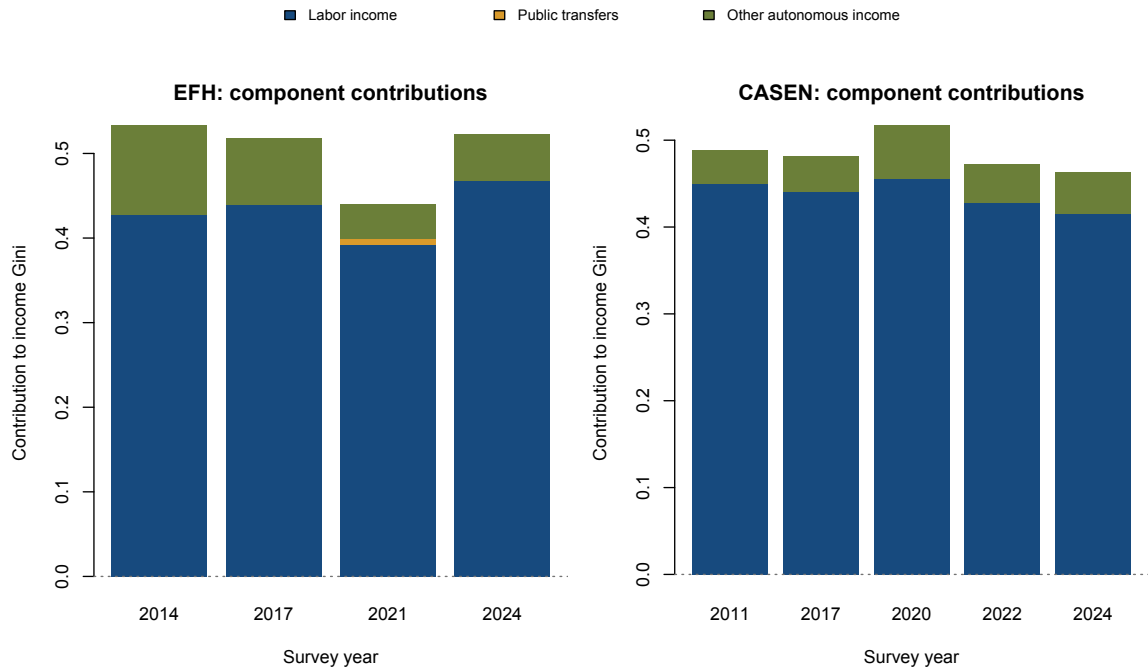


Figure 4: Income-component contributions to the Gini

Public transfers’ share of measured EFH effective income rose from 0.8% of EFH effective

income in 2017 to 13.5% in 2021, then fell to 3.1% in 2024. The 2017 to 2021 income-Gini decline is accounted for by contribution changes of -0.050 for labor income, -0.039 for other autonomous income, and 0.009 for transfers. Transfers' larger share of measured income displaced more concentrated income shares, while transfer receipts became less concentrated among households ranked near the bottom of total income, offsetting part of that compression. CASEN follows a different pandemic sequence: its income Gini rises in 2020, falls sharply by 2022, and edges down again by 2024. Component contributions are exact descriptive accounts and do not identify transfer effects.

The exact decomposition helps avoid a common misreading of transfer income. A component can reduce the total Gini through two distinct margins. Its share of mean income may rise while more concentrated components lose share, and its own concentration coefficient may be low or negative because it is received disproportionately by households below the total-income mean. These margins need not point in the same direction. In the 2021 EFH, transfers account for a much larger part of mean income, yet their positive contribution partly offsets rather than mechanically explains the entire compression. Most of the fall is recorded in the smaller labor and other-autonomous contributions.

Even this exactness has a narrow meaning. A smaller labor contribution can reflect lower labor's mean share, less dispersion in labor income, reranking when transfers are included in total income, or all three. It does not by itself say that labor-market inequality improved. Likewise, a negative CASEN transfer contribution says that transfer receipts are concentrated toward lower total-income ranks; it does not estimate the Gini that would prevail if the transfer system were removed and households did not change behavior.

CASEN tells a different component story. Its Gini rises in 2020 because the labor and other-autonomous contributions increase more than the transfer contribution becomes negative. By 2022 the labor contribution falls sharply, and it declines again by 2024. The difference matters for interpretation: the two surveys do not merely attach different levels to a common turning point. They record different timing and different component contributions.

The common endpoint sharpens that contrast. Labor income adds 0.031 to the EFH Gini between 2017 and 2024 but subtracts 0.021 from the CASEN Gini. Other autonomous income partly offsets the difference, while transfers contribute about  $-0.004$  in each survey. The new covariance-agnostic component intervals, reported in the supplement, are wide enough to caution against treating any one component as an isolated mechanism. The account identifies where the two observed changes reside; it cannot say whether the labor contrast reflects economic coverage, reporting, correction, or sample composition.

The decomposition is deliberately silent about statistical significance for individual mechanisms. It reports shares, concentration coefficients, and exact contributions with source-specific uncertainty, but it does not search across component-level hypotheses. The key conclusion is narrower. The EFH 2021 compression coincides with a temporary income composition that is absent by 2024, while CASEN's higher-frequency path places its pandemic peak in 2020 and its

decline thereafter.

The timing difference also disciplines the interpretation of survey mode. Both surveys had unusual pandemic collection, yet the highest recent CASEN Gini is in 2020 while the lowest recent EFH Gini is in 2021. Telephone collection alone therefore cannot explain the direction of both patterns without a model of who responded, what was asked, and the economic conditions at each field date. The intermediate CASEN rounds are therefore valuable as a description of its own path, not as a mode correction to EFH.

### 5.3 The survivor-domain check

The repeated cross-section could reverse because the 2024 refresh sample changed who was observed. Table 3 tests whether that explanation is sufficient by holding the linked household set fixed and applying a common weighting system to both years.

| Domain                         | Period              | Income Gini or change |
|--------------------------------|---------------------|-----------------------|
| Repeated cross-section         | 2021                | 0.439 [0.427, 0.451]  |
| Repeated cross-section         | 2024                | 0.523 [0.494, 0.552]  |
| 2024 survivor domain           | 2021                | 0.444 [0.425, 0.462]  |
| 2024 survivor domain           | 2024                | 0.528 [0.469, 0.587]  |
| Repeated cross-section         | 2021 to 2024 change | 0.084 [0.043, 0.125]  |
| Survivors, common 2024 weights | 2021 to 2024 change | 0.084 [0.007, 0.162]  |
| Survivors, common 2021 weights | 2021 to 2024 change | 0.096 [0.022, 0.170]  |

Table 3: Repeated cross-section and survivor-domain income Ginis

Level estimates use the 2024 final weight and all 1,000 provider replicates for both years; the change is repeated with common 2021 weights and replicates. Change intervals use the covariance-agnostic Cauchy bound because cross-wave imputations were not jointly drawn. This is a 2024 survivor domain, not a population panel.

Among the 1,801 households reinterviewed in 2024, the common-weight income Gini rose from 0.444 to 0.528, a change of 0.084 [0.007, 0.162]. With common 2021 weights the corresponding change is 0.096 [0.022, 0.170]. The survivor-domain reversal means that refresh-sample composition is not sufficient to account for the repeated-cross-section result. This domain is not a population panel: survival, nonresponse, and separately drawn cross-wave imputations limit its interpretation.

The attrition diagnostic is reassuring but not dispositive. Using 2021 weights, survivors and the non-reinterviewed remainder of the 2021 refresh cohort have similar average baseline income, wealth, household size, macrozone shares, and income-stratum shares. The standardized differences for income, wealth, and household size are 0.00, 0.04, and 0.09 in absolute value. That comparison cannot rule out selection on unobserved outcomes or future changes, but it supports the limited inference that refresh composition by itself does not generate the reversal. The full continuous and categorical balance table appears in the supplement.

The 2024 final weights describe the domain observed again in 2024 and are therefore applied to both dates. As a check, the same households are also estimated at both dates with common

2021 weights. Both versions preserve the reversal and have conservative intervals above zero. Neither exercise estimates the 2021 population: each describes the eventual survivor set under one fixed weighting system.

The exercise isolates one explanation. Because the increase remains after fixing the household set and weighting system, newly sampled households cannot account for it by themselves. Attrition, measurement changes within surviving households, and genuine changes in their incomes remain possible.

## 5.4 What the diagnostics leave unresolved

The component and survivor exercises narrow the space of explanations but do not partition the EFH–CASEN gap. The direct comparison fixes the survey year, urban population, household unit, and treatment of imputed rent, so those choices do not explain the 2024 disagreement. Survey purpose, respondent selection, recall periods, question order, income editing and correction, weighting, nonresponse, upper-tail coverage, and fieldwork mode remain jointly different. The released files cannot vary one of these margins while holding the others fixed. Attributing the remainder would require a common-microdata bridge or administrative linkage.

## 6 Discussion and conclusion

Did inequality fall in Chile? Historically, the CASEN record says yes. It places income inequality well below its 1990s level by the middle of the 2010s, and several international databases reproduce that decline after applying different methods to CASEN. Because EFH begins in 2007, it cannot adjudicate the earlier fall. It can show how far the conclusion travels into the 2020s, across survey systems, and from income to wealth.

On those margins, the answer is less uniform. CASEN’s closest income Gini is lower in 2024 than in 2017. EFH instead falls to an exceptional 2021 trough and returns to approximately its 2017 level by 2024. Its 2007–2024 endpoint leaves room for either modest net compression or little net movement, but it does not show a continuing post-2017 decline. Assigning income to represented persons or including imputed rent preserves the directional contrast. The direct difference in 2017–2024 changes remains imprecise, so the paper establishes different observed paths and growing disagreement rather than a statistically resolved ranking of survey trends.

The component and survivor exercises make that disagreement more informative. The 2021 EFH trough coincides with a temporary composition of measured income: public transfers take a much larger share, while labor and other-autonomous contributions contract. By 2024 that composition has receded. The income rebound also remains among the 1,801 reinterviewed households under either common weighting system, and those survivors look similar at baseline to the non-reinterviewed refresh remainder on the observed diagnostics. Sample refresh alone is therefore insufficient, although economic change, attrition, questionnaire design, nonresponse,

mode, and within-household measurement remain entangled.

Wealth measures a stock and necessarily answers a different question from monthly income; the EFH concept also excludes mandatory pensions, macro scaling, and a corrected extreme upper tail. Even within that bounded concept, EFH does not support a single story of declining measured inequality through 2024: signed non-pension wealth inequality rose between 2021 and 2024, and the increase survives share, zero-cap, p99, and p99.5 influence checks.

The assessment therefore depends on the period, resource, statistical unit, construction method, and underlying microdata. The closest comparison holds year, urban coverage, household unit, and imputed rent fixed, yet it cannot identify which remaining survey difference produces the gap. Source lineage is consequently part of substantive inference: agreement among CASEN descendants demonstrates robustness across treatments of common microdata, not independent replication across survey systems. Establishing what lies behind the disagreement will require a bridge design or common administrative benchmark. That distinction matters for evaluations of distributional progress: several published indicators can agree without adding independent evidence about the population trend. Chile’s historical decline remains well supported, but the available evidence does not justify extending it unconditionally through 2024 across income and wealth.

## Declarations

### Data and code availability

EFH microdata are restricted and are not redistributed. Researchers must obtain them from Banco Central de Chile and comply with the provider’s terms. CASEN inputs are frozen official public releases with checksums and retrieval metadata. The repository contains reproducible estimators, symbolic and numerical checks, Monte Carlo validation, output contracts, and a fail-closed build, and is available on [GitHub](#). It is not a self-contained public replication package because the EFH microdata and unrestricted dissemination rights are not included.

### Conflicts of interest

The author reports no conflict of interest.

## References

Facundo Alvaredo, Mauricio De Rosa, Ignacio Flores, and Marc Morgan. The inequality (or the growth) we measure: Data gaps and the distribution of incomes. Working Paper 2022/07, World Inequality Lab, 2022.

Facundo Alvaredo, François Bourguignon, Francisco H. G. Ferreira, and Nora Lustig. Inequality

- bands: Seventy-five years of measuring income inequality in latin america. *Oxford Open Economics*, 4(Supplement 1):i9–i35, 2025. doi: 10.1093/ooec/odae018. Published version checked 2 September 2026; Figure 4 and Table 2 motivate the Chile source-lineage audit.
- Anthony B. Atkinson and Andrea Brandolini. Promise and pitfalls in the use of “secondary” datasets: Income inequality in OECD countries as a case study. *Journal of Economic Literature*, 39(3):771–799, 2001. doi: 10.1257/jel.39.3.771.
- Banco Central de Chile. Encuesta Financiera de Hogares (EFH): Restricted microdata and accompanying documentation, 2007–2024. Survey waves used under the provider’s applicable access conditions; microdata are not redistributed with this repository.
- Thomas Blanchet, Ignacio Flores, and Marc Morgan. The weight of the rich: Improving surveys using tax data. *The Journal of Economic Inequality*, 20(1):119–150, 2022.
- Bastián Castro Nofal, Ignacio Flores, and Pablo Gutiérrez Cubillos. From housing gains to pension losses: Micro–macro integration to reveal wealth inequality dynamics in chile. *Journal of Public Economics*, 259:105665, 2026. doi: 10.1016/j.jpubeco.2026.105665.
- CEDLAS and World Bank. SEDLAC: Methodology and household-survey inventory, 2025. URL <https://www.cedlas.econo.unlp.edu.ar/wp/en/estadisticas/sedlac/metodologia-sedlac/>. August 2025 vintage; retrieved 2 September 2026.
- Lidia Ceriani and Paolo Verme. A world of ginis, 2026. URL <https://arxiv.org/abs/2607.24175>. Working paper dated 24 July 2026; retrieved 3 September 2026.
- Enzo A. Cerletti, Magdalena Cortina, Alejandra Inzunza, Felipe Martínez, and Patricio Toro. The finances of chilean households during the pandemic: An assessment from the 2021 household financial survey. Working Paper 1008, Banco Central de Chile, 2024. URL <https://www.bcentral.cl/web/banco-central/contenido/-/detalle/documento-de-trabajo-n-1008>.
- Paulo Cox, Eric Parrado, and Jaime Ruiz-Tagle. The distribution of assets, debt and income among chilean households. Working Paper 388, Banco Central de Chile, 2006. URL [https://www.bcentral.cl/documents/33528/133326/DTBC\\_388.pdf](https://www.bcentral.cl/documents/33528/133326/DTBC_388.pdf).
- Economic Commission for Latin America and the Caribbean. Conceptual note: Indicators in cepalstat estimated from the household survey database (badehog). Technical report, United Nations, 2025. URL <https://hdl.handle.net/11362/81902>. June 2025 methodological note; retrieved 2 September 2026.
- Tasha Fairfield and Michel Jorratt De Luis. Top income shares, business profits, and effective tax rates in contemporary Chile. *Review of Income and Wealth*, 62(S1):S120–S144, 2016. doi: 10.1111/roiw.12196.

- Ignacio Flores, Claudia Sanhueza, Jorge Atria, and Ricardo Mayer. Top incomes in Chile: A historical perspective on income inequality, 1964–2017. *Review of Income and Wealth*, 66(4): 850–874, 2020. doi: 10.1111/roiw.12441.
- Instituto Nacional de Estadísticas. Encuesta de presupuestos familiares: Información de la encuesta, 2022. URL <https://www.ine.gob.cl/estadisticas-por-tema/sociedad-y-condiciones-de-vida/encuesta-de-presupuestos-familiares/informacion-de-la-encuesta>. Official survey documentation, accessed 2 September 2026.
- Instituto Nacional de Estadísticas. Methodology of the supplementary survey of income. Technical report, Instituto Nacional de Estadísticas de Chile, 2025. URL <https://www.ine.gob.cl/docs/default-source/supplemental-survey-on-income/methodologies/documents/methodology-supplementary-survey-of-income-esi.pdf>. Methodological document retrieved 2 September 2026.
- Osvaldo Larrañaga, Benjamín Echeopar, and Nicolás Grau. Una nueva estimación de la desigualdad de ingresos en Chile. *Estudios Públicos*, (167):45–76, 2022. doi: 10.38178/07183089/1229210914.
- LIS Cross-National Data Center. Chile: A new member in the lis database, 2018. URL <https://www.lisdatacenter.org/wp-content/uploads/files/nl-2018-6.pdf>. Identifies the Chilean LIS income datasets as CASEN-based; retrieved 2 September 2026.
- LIS Cross-National Data Center. Luxembourg wealth study: Chile data release and revisions, 2025. URL <https://www.lisdatacenter.org/news-and-events/2025-winter-data-splash/>. Identifies CL07, CL14, CL17, CL21, and CL24 as EFH-based; retrieved 2 September 2026.
- Felipe Martínez and Francisca Uribe. Distribución de riqueza no previsional de los hogares chilenos. Documento de Trabajo 806, Banco Central de Chile, 2017. URL [https://www.bcentral.cl/documents/33528/133326/DTBC\\_806.pdf](https://www.bcentral.cl/documents/33528/133326/DTBC_806.pdf).
- Ministerio de Desarrollo Social y Familia. Encuesta de Caracterización Socioeconómica Nacional (CASEN): Microdata and methodological documentation, 2011–2024. Official public microdata waves 2011, 2017, 2020, 2022, and 2024; exact retrieval dates and checksums are recorded in the project manifest.
- Ministerio de Desarrollo Social y Familia. Encuesta Casen en Pandemia 2020: Documentation and public archive, 2022. URL <https://observatorio.ministeriodesarrollosocial.gob.cl/encuesta-casen-en-pandemia-2020>. Revised public documentation records a sequential mixed mode dominated by telephone interviewing and cautions about comparisons across rounds.
- Ministerio de Desarrollo Social y Familia. Informe sobre desigualdad de ingresos en Chile. Technical report, Gobierno de Chile, 2026. URL [https://observatorio.ministeriodesarrollosocial.gob.cl/storage/docs/desigualdad/Informe\\_Desigualdad\\_2026.pdf](https://observatorio.ministeriodesarrollosocial.gob.cl/storage/docs/desigualdad/Informe_Desigualdad_2026.pdf). March 2026; retrieved 2 September 2026.

- OECD. Wealth distribution database: Main characteristics of data sources, 2025. URL <https://www.oecd.org/content/dam/oecd/en/data/datasets/income-and-wealth-distribution-databases/wdd-metadata-sources-and-years.pdf>. Identifies the Survey of Household Finances and Central Bank of Chile as the Chilean source; retrieved 2 September 2026.
- OECD. Metadata on the oecd income distribution database: Sources and income years, 2026. URL <https://www.oecd.org/content/dam/oecd/en/data/datasets/income-and-wealth-distribution-databases/idd-metadata-sources-and-income-years.pdf>. June 2026 version; identifies CASEN as the Chilean source and equivalized disposable cash income as the resource.
- Francisco Parro and Loreto Reyes. The rise and fall of income inequality in Chile. *Latin American Economic Review*, 26(3), 2017. doi: 10.1007/s40503-017-0040-y.
- Thomas Piketty, Emmanuel Saez, and Gabriel Zucman. Distributional national accounts: Methods and estimates for the united states. *The Quarterly Journal of Economics*, 133(2): 553–609, 2018.
- Daniel Rodríguez Tapia. ¿sigue rindiendo estudiar? educación, ingresos laborales y desigualdad en chile, 2006–2024. *Políticas Públicas*, 18(2), 2026. doi: 10.35588/y1ht4559. URL <https://revistas.usach.cl/ojs/index.php/politicas/article/view/8159>.
- Claudia Sanhueza and Ricardo Mayer. Top incomes in Chile using 50 years of household surveys: 1957–2007. *Estudios de Economía*, 38(1):169–193, 2011. doi: 10.4067/S0718-52862011000100007.
- Frederick Solt. Measuring income inequality across countries and over time: The standardized world income inequality database. *Social Science Quarterly*, 101(3):1183–1199, 2020. doi: 10.1111/ssqu.12795. SWIID Version 9.92 metadata checked 2 September 2026.
- Miguel Székely and Marianne Hilgert. What’s behind the inequality we measure? an investigation using latin american data. *Oxford Development Studies*, 35(2):197–217, 2007. doi: 10.1080/13600810701427626.
- UNU-WIDER. World income inequality database, 2025. URL <https://www.wider.unu.edu/database/world-income-inequality-database-wiid>. Version 29 April 2025; the temporarily withdrawn 2026 update is not used.
- World Bank. March 2023 update to the poverty and inequality platform: What’s new. Technical report, World Bank, 2023. URL <https://documents1.worldbank.org/curated/en/099923403272329672/pdf/IDU089370bcb048b9044fd0ab49037249b87aef6.pdf>. Documents the use of SEDLAC-harmonized Latin American surveys in PIP; retrieved 2 September 2026.
- World Inequality Lab. World inequality report 2026: Country sheets. Technical report, World Inequality Lab, 2026. URL [https://wir2026.wid.world/www-site/uploads/2025/11/WIR26\\_Country\\_Sheets.pdf](https://wir2026.wid.world/www-site/uploads/2025/11/WIR26_Country_Sheets.pdf). Chile country sheet; retrieved 3 September 2026.